
Failure Modes in Causal Steering of Protein Language Models: Why Proxy Accuracy Does Not Guarantee Steering Success

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Abstract

1 Difference-of-means activation steering has reproduced a known thermostability
2 effect in ESM2-650M. We test whether the same method transfers to five new
3 protein properties and whether a more accurate scoring proxy provides a better
4 steering target. The binding-affinity proxy has the highest held-out correlation in
5 the study ($r = 0.80$), but its steering direction has no measurable effect. A weaker
6 catalytic-activity proxy ($r = 0.22$) produces a consistent effect across three random
7 seeds. An immunogenicity proxy that initially appears predictive is explained by
8 source-organism confounding, so that target does not reach the steering stage.
9 An expression-yield effect fails a residue-exclusion check and overlaps with the
10 steering direction for intrinsic disorder. The evaluation also exposes two problems
11 in our own analysis. One comparison used an intervention strength at which a
12 baseline had already collapsed, while one robustness criterion changed its verdict
13 across seeds. These cases show why proxy validation and a single successful run
14 are not sufficient evidence for a reliable steering intervention.

15 1 Problem

16 1.1 Research question and setting

17 Difference-of-means activation steering adds a direction to a protein language model’s residual stream
18 during generation. The direction is formed from the difference between mean activations for examples
19 with high and low values of a target property. The method has reproduced a known thermostability
20 effect in ESM2-650M [Huang et al., 2025].

21 Applying the method to a new property raises two separate questions. A scoring proxy may agree
22 with experimental labels, but the steering direction built for that property may still be ineffective. A
23 result may also depend on the train-test split, random control direction, or intervention strength. We
24 therefore evaluate both the steering direction and the stability of the evidence used to support it.

25 1.2 Targets and measurements

26 The study covers five properties that were not part of the original thermostability experiment: binding
27 affinity, catalytic activity, intrinsic disorder, immunogenicity, and expression yield. For each property,
28 we first measure the Pearson correlation between a compositional proxy and experimental labels on
29 held-out data. If the proxy passes this check, we compare the property-derived steering direction with
30 a random direction of the same norm.

31 The main outcome is the score difference between the learned and random directions. We also test
32 whether the effect changes monotonically with intervention strength, remains after dominant residue
33 substitutions are excluded from scoring, and repeats across random seeds. If proxy accuracy were a
34 useful way to select steering targets, properties with stronger held-out correlations would also have
35 larger or more consistent steering effects. We do not observe that pattern.

36 **2 Proposed Approach**

37 **2.1 Steering procedure**

38 For each target, we divide real protein sequences into high-property and low-property groups. We
39 compute the mean activation for each group at every layer of ESM2-650M and subtract the two
40 means. During generation, we add a scaled version of this direction to the residual stream. The same
41 code handles vector construction, random controls, sequence generation, bootstrap testing, and output
42 filtering for every target. Using one implementation allows us to change the dataset and scoring proxy
43 without changing the intervention or evaluation procedure.

44 **2.2 Six evaluation requirements**

45 We fix the following requirements before testing the new properties:

- 46 1. The property-derived direction must outperform a random direction of the same norm in a
47 paired bootstrap test with 10,000 resamples.
- 48 2. The effect must increase monotonically across at least three intervention strengths in a range
49 where generation remains stable.
- 50 3. The effect must remain after the most frequently substituted residues are excluded from the
51 score.
- 52 4. The scoring proxy must agree with experimental labels on held-out data before steering is
53 attempted.
- 54 5. The method must outperform an existing baseline when a suitable baseline is available.
- 55 6. An evaluation of a new property must contain at least 150 held-out sequences.

56 The third requirement checks for compositional collapse. A steering direction can appear successful
57 if it repeatedly inserts one residue that directly increases the proxy score. Excluding the dominant
58 substitutions tests whether the effect extends beyond that narrow change. We use three assessment
59 outcomes. If proxy validation fails, we reject the proxy and do not attempt steering. If the main
60 steering test succeeds but residue exclusion fails or changes across seeds, we classify the evidence as
61 inconclusive. We call a result supported only when it satisfies every applicable requirement.

62 **2.3 Attention-head comparison**

63 We also examine whether the same problem appears at the level of individual attention heads.
64 Following Vig et al. [2021], we measure contact enrichment in their protein language model and
65 select the most enriched head in our data. We ablate that head and compare the change in masked-
66 residue prediction with a low-correlation control head. We then repeat the ablation for all 480 heads
67 in the model.

68 **3 Observed Outcome**

69 **3.1 Proxy accuracy does not predict steering effects**

70 Table 1 summarizes the five target properties. Binding affinity has the strongest proxy validation
71 but no measurable steering effect. Catalytic activity has the weakest proxy among the four runnable
72 targets but passes every requirement on three random seeds.

Table 1: Results for the five target properties. Proxy r is the Pearson correlation with experimental labels on held-out data, measured before steering.

Target	Dataset	Proxy r	Steering result	Assessment
Binding affinity	ProteinGym DMS (KRAS/DARPin)	0.80	no measurable effect	Unsupported
Catalytic activity	DLKcat (cross-protein)	0.22	repeats across 3 seeds	Supported
Intrinsic disorder	DisProt	0.45	main effect 3/3; check 2/3	Inconclusive
Immunogenicity	IEDB (4-tier)	≤ 0.10	steering not attempted	Proxy rejected
Expression yield	eSol	0.31	fails residue-exclusion check	Inconclusive

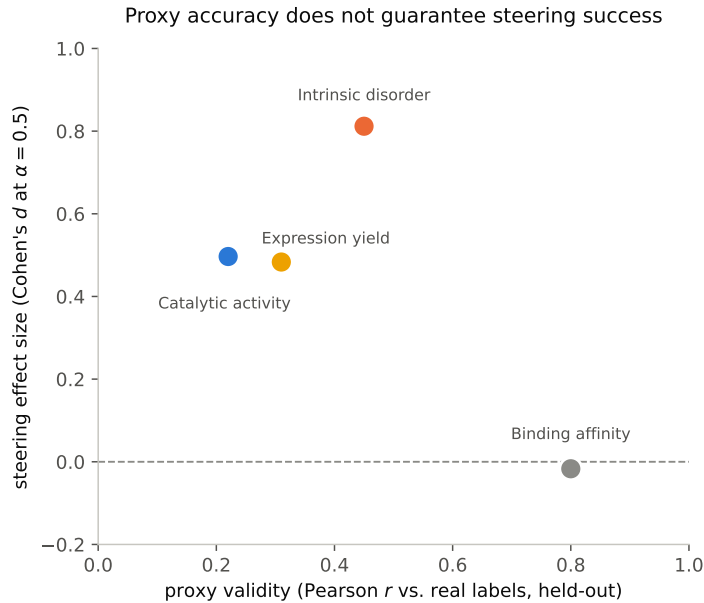


Figure 1: Proxy correlation and steering effect size for the four targets that reach the steering stage. Effect size is Cohen’s d at $\alpha = 0.5$ for the learned direction relative to the matched random direction. The four points show no positive relation between the two measurements.

73 3.2 Binding affinity

74 We use a ProteinGym deep-mutational-scanning assay [Notin et al., 2023] of KRAS binding to a
75 DARPin. The proxy uses position-specific mutational-sensitivity weights learned on the training split
76 and reaches $r = 0.80$ on held-out data. A weight-shuffle control gives $r = -0.001$. The proxy also
77 generalizes to positions excluded from weight fitting ($r = 0.54$) and from single to double mutants
78 ($r = 0.70$).

79 The steering direction nevertheless has no detectable effect at any tested strength. Because generation
80 remains stable and produces no degenerate sequences at those strengths, output collapse does not
81 explain the result.

82 3.3 Catalytic activity

83 For catalytic activity, we use a cross-protein enzyme turnover dataset from DLKcat [Li et al., 2022].
84 The proxy measures the fraction of glycine minus the fraction of arginine. Its held-out correlation
85 is $r = 0.22$, the weakest among the four targets that reach the steering stage. Even so, the steering
86 effect is significant and monotonic, remains after residue exclusion, and repeats across three random
87 seeds.

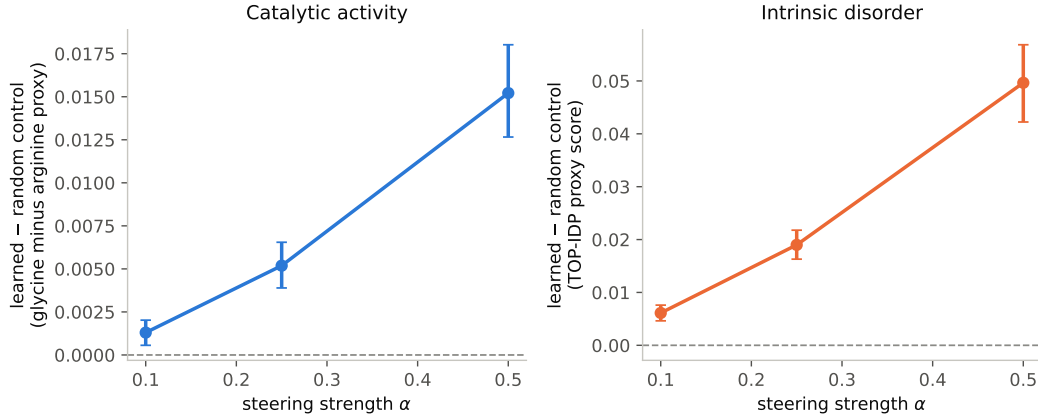


Figure 2: Dose-response curves for catalytic activity and intrinsic disorder. Both effects increase monotonically across the tested intervention strengths. Error bars show 95% bootstrap confidence intervals.

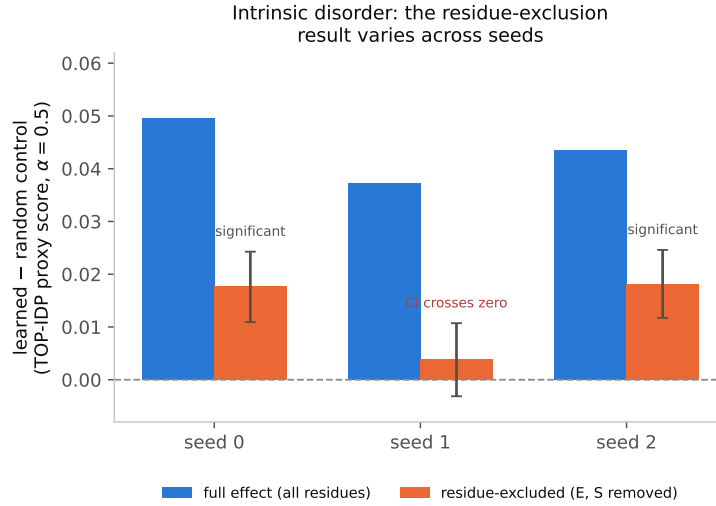


Figure 3: Intrinsic-disorder effects before and after residue exclusion for three random seeds. The main effect is stable, but the confidence interval for the residue-excluded score crosses zero on seed 1.

88 3.4 Intrinsic disorder

89 We use DisProt annotations [Quaglia et al., 2022]. The intrinsic-disorder proxy uses the published
 90 TOP-IDP amino-acid scale [Campen et al., 2008]. It reaches held-out $r = 0.45$ and per-residue
 91 AUC 0.71. The main steering effect is significant and monotonic on all three random seeds. The
 92 residue-exclusion check passes on two seeds but fails on the third. The effect is repeatable, but the
 93 evidence against a simple compositional artifact is not equally strong on every seed.

94 3.5 Immunogenicity

95 For immunogenicity, we use records from the Immune Epitope Database [Vita et al., 2019]. Candidate
 96 proxies correlate with a surrogate binding-affinity endpoint but fall to $r \leq 0.10$ on the measured
 97 T-cell-response endpoint. One proxy for full-length antigens initially reaches $r = 0.38$. Further
 98 testing shows that organism identity explains 58% of this signal. When organism identity is held out
 99 during cross-validation, the correlation changes to -0.32 . Because no proxy passes the validation
 100 requirement, we do not run a steering experiment.

101 3.6 Expression yield

102 We use expression-yield measurements from eSol [Niwa et al., 2009]. The proxy reaches $r = 0.31$ –
103 0.34 and produces a significant dose response. The effect disappears when the dominant residue
104 substitutions are excluded. The main result is therefore inconclusive under the fixed protocol.

105 3.7 Attention-head ablation

106 In our replication of Vig et al. [2021], the most contact-enriched head has 12.9-fold enrichment over
107 background. Ablating it produces a change that is statistically indistinguishable from ablating a
108 low-correlation control head. The full sweep does not provide a precise rank because the evaluation
109 uses 104 sampled positions and many heads have identical measured effects. The selected head
110 shares its value with 131 other heads, forming a tied block that spans ranks 286–417. At the same
111 resolution, 166 heads show no measurable ablation effect. Strong contact correlation therefore does
112 not imply a large or unique contribution to masked-residue prediction.

113 4 Reason for Failure

114 4.1 The binding-affinity groups differ at few residues

115 The binding-affinity sequences are point mutants of one 188-residue reference protein. The high-
116 and low-binding groups differ by only one or two residues per sequence. Their mean per-residue
117 compositional distance is 0.003, compared with 0.02–0.05 for the three cross-protein datasets.

118 This difference offers a possible explanation for the failed intervention. The proxy uses position-
119 specific weights, so it can predict binding labels from a small number of mutations. In contrast, the
120 steering direction averages activations over groups of nearly identical sequences. The few mutations
121 that distinguish the groups may contribute too little to that average. Our evidence supports this
122 explanation as a possibility, not as a demonstrated mechanism.

123 4.2 The expression-yield direction overlaps with disorder

124 We compare the expression-yield steering vector with every other target vector. Its cosine similarity
125 with the intrinsic-disorder direction is +0.38 across all 33 layers and rises to +0.56–0.67 in the
126 three deepest layers. This overlap shows that the directions are not orthogonal and may capture
127 some of the same compositional variation. It does not show that disorder causes the expression-yield
128 result. Because the expression-yield result also fails residue exclusion, we do not count it as separate
129 evidence for a new steering effect. Vector similarity is inexpensive to compute and is useful when
130 several related targets are studied in the same model.

131 4.3 Unstable generation made five-layer steering look stronger

132 In a follow-up thermostability experiment, we compared steering all 33 layers with steering only five
133 layers previously identified as important. An early analysis selected the best intervention strength
134 from the full tested range. At that strength, generation under the 33-layer intervention had already
135 collapsed, while generation under the 5-layer intervention had not. The 5-layer version appeared
136 better only because it failed at a higher strength.

137 We restricted the comparison to a range where both interventions produced stable sequences and reran
138 the experiment. The 5-layer intervention then retained 43–59% of the full effect. This is a useful
139 partial result, but it does not support the original claim that five layers perform as well as all 33.

140 4.4 A robustness criterion changed across seeds

141 The intrinsic-disorder effect repeats across all three seeds, but its residue-exclusion check passes on
142 only two. A single run could therefore support either a pass or a failure for that criterion. Repeating
143 only the main effect is not enough when a separate artifact check is used to justify the result.

144 4.5 Practical recommendations

145 Before steering, researchers should measure how much the high- and low-property groups differ,
146 not only how well the scoring proxy predicts labels. Comparisons between methods or layer sets
147 should use a range in which every intervention produces stable output. Artifact checks should be
148 repeated across seeds when they support an important claim. When several properties are studied
149 together, similarities between their steering vectors can reveal that two apparent effects may not be
150 independent.

151 5 Related Work

152 Huang et al. [2025] introduced difference-of-means activation steering for protein language models
153 and validated the method on thermostability. We apply the same intervention to five new properties
154 under one fixed evaluation protocol.

155 Vig et al. [2021] identified attention heads whose attention patterns correlate with 3D protein contacts.
156 We reproduce their analysis, test the most enriched head in our data by causal ablation, and extend
157 the analysis to all heads in the model.

158 Adams et al. [2025] used linear probes to identify biological information in sparse-autoencoder
159 features from ESM-2. Their steering experiment tested sequence coherence rather than agreement
160 with an experimental property label. Our study asks whether correlational evidence predicts the
161 outcome of a downstream intervention.

162 6 Limitations

163 All experiments use ESM2-650M at one model scale. The proxies are compositional heuristics
164 rather than learned predictors. A stronger non-compositional proxy could behave differently. We
165 selected each final proxy from several candidates using the same held-out split on which we report its
166 correlation, so the reported correlations are likely optimistic.

167 We ran multiple significance tests without a formal correction. Agreement across intervention
168 strengths and seeds reduces this risk but does not remove it. Seed robustness was checked for only
169 two targets. We also did not evaluate generated sequences with a wet-lab assay or a structural
170 plausibility model. A successful protocol result means that the intervention changes a validated proxy.
171 It does not establish that the generated protein has the intended biological property.

172 7 Conclusion

173 Across these experiments, higher proxy accuracy does not guarantee more successful steering. The
174 binding-affinity result shows that a well-validated score can still produce an ineffective difference-
175 of-means direction. The seed and intervention-range checks also show how a plausible result can
176 receive the wrong interpretation. Reliable steering claims require direct tests of the intervention, its
177 artifacts, and its stability.

178 LLM usage disclosure

179 Claude (Anthropic) assisted with code writing, experiment scripts, literature searches, and manuscript
180 drafting. We chose the study design, ran and checked the experiments, and verified each numerical
181 claim against committed outputs.

182 Code and data availability

183 Code, data-processing scripts, and saved results will be released after the double-blind review period.
184 The submission does not link to an identifying repository.

Reproducibility statement

All five targets use the same implementation for steering-vector construction, matched-norm random controls, paired bootstrap testing, and output filtering. The multi-seed experiments rerun the complete pipeline rather than resampling one run’s output. Saved result files support the numerical claims reported in the paper.

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